YASIR HUSSAIN

Avionics Engineer

UAV & VTOL Systems | AI & Automation | Data & Power Platform

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LinkedIn | GitHub | Kaggle | Portfolio

PROFESSIONAL SUMMARY

Results-driven Avionics Engineer experienced in UAV avionics integration, flight-controller configuration, sensor calibration, and fail-safe autonomous systems across multirotor, fixed-wing, and VTOL platforms using ArduPilot, Mission Planner, and Betaflight. Combines avionics expertise with 2+ years in Python, machine learning, LLM automation, and enterprise data and Power Platform solutions, applying AI to telemetry analysis, mission automation, and reliable real-world autonomous systems. Based in Saudi Arabia and aligned with Vision 2030 priorities.

TECHNICAL SKILLS

Avionics & UAV: ArduPilot (QuadPlane/Copter), Mission Planner, QGroundControl, Betaflight 4.5 & RPM Notch, Pixhawk 6C / Cube Orange+, MAVLink Telemetry, RTK GNSS, CAN / UART / I2C, Arduino, ESP32, STM32

AI, ML & Automation: Python, TensorFlow, PyTorch, Scikit-learn, NLP, Computer Vision (OpenCV, YOLOv8), LangChain, LangGraph, RAG, LLM Agents, Power Automate, n8n, Zapier

Data & BI: SQL/MySQL, ETL/ELT, Power BI, DAX, Tableau, Microsoft Fabric, Dataverse, MongoDB

Development & Platforms: Python, TypeScript, JavaScript, C++ / Embedded C, React, Flask, FastAPI, Docker, Git/GitHub Actions, Azure, SPFx, PCF, Power Apps, SharePoint

PROFESSIONAL EXPERIENCE

Avionics Engineer — Apr 2026 – Present

Beta Industrial • *Saudi Arabia*

- Lead UAV avionics integration and flight-controller tuning (ArduPilot/Betaflight) across multirotor, fixed-wing, and VTOL platforms using Arduino, ESP32, STM32, Pixhawk 6C Mini, and Orange Cube.
- Calibrate flight sensors including RTK GNSS, Pitot-static airspeed (MS4525DO), and dual-redundant CAN/internal compasses; configure fail-safes such as automated Return-to-Launch (RTL).
- Configure and tune flight parameters with Betaflight and Mission Planner, and troubleshoot hardware-software integration for autonomous flight systems.

Data Scientist / Power Platform Developer — Nov 2024 – Mar 2026

Numrex • *Lahore, Pakistan*

- Delivered a Power BI Business Performance Dashboard for Associated British Foods (ABF) covering sales, supply chain, production, and retail performance across five divisions.
- Built a secure HR Power App for St. James's Place (Power Apps & Power Automate) with data model, HR screens, onboarding automation flows, and role-based security.
- Designed a Power BI timesheet dashboard with custom DAX measures, improving operational efficiency by 30%.
- Built CRM forms with Power Apps and Dataverse validation rules; developed custom SPFx web parts (100% client satisfaction) and automated document library expiry tracking with Power Automate.

Power Platform & SharePoint Developer — Nov 2024 – Dec 2025

KeaPoint (UK) • *Hybrid*

- Delivered SharePoint modernization and Power Platform solutions for UK clients using SPFx, Power Apps, Power Automate, and Dataverse data modeling.

Model Training (Content Analyst) — May 2025 – Aug 2025

Turing (Meta) • *Remote*

- Evaluated and fine-tuned large language models with alignment data, prompt engineering, and RLHF scoring; annotated 120+ video clips for multimodal ML training, improving model accuracy by 15% and cutting labeling time by 20%.

Power BI Dashboard Developer
Jun 2024 – Aug 2024

Activision • *Bahawalpur, Pakistan*

- Built Power BI dashboards with DAX and data modeling on large gaming and Superstore datasets, reducing reporting time by 30%.

Data Science Intern
May 2023 – Aug 2023

CodSoft • *Remote*

- Built ML predictive models and data-cleaning pipelines in Python; improved data processing speed by 30% across 4 machine learning projects.

KEY PROJECTS

- **Autonomous Hybrid VTOL Mapping & Telemetry Platform** *(ArduPilot QuadPlane, Mission Planner, MAVLink, RTK-GNSS).* Configured, wired, and flight-tested an autonomous VTOL UAV with waypoint navigation, airspeed calibration, and fail-safe RTL; 60+ minute autonomous cruise.
- **Drone Telemetry ETL & Anomaly Scoring** *(Python, Pandas, Scikit-learn, pymavlink).* Parses DFLog/tlog telemetry and uses Isolation Forest to flag IMU vibration and voltage anomalies before hardware failure.
- **Legal Chatbot for KPK Police** *(LangChain, OpenAI API, Pinecone, FastAPI).* Production RAG chatbot for instant legal statute lookup, reducing query resolution time by 50%.
- **Bilingual AI Chatbot (AR/EN)** *(Python, Flask, LangChain, Groq API, LLaMA 3.3-70B).* RAG chatbot with RTL Arabic UI and voice input, deployed live for retail/logistics.
- **AI CV & ATS Score Analyzer** *(Python, Flask, NLP).* Extracts 100+ skills, matches job descriptions, and returns ATS scores; deployed live for the Saudi market.
- **ZATCA-Compliant VAT Invoice Generator** *(Python, Flask, ReportLab).* Generates Phase 1 compliant invoices with QR codes and Arabic/English toggle in under 10 seconds.
- **Security System for Bike Entry** *(YOLOv8, OpenCV, Flask).* Real-time computer vision entry detection with 91% accuracy.

EDUCATION

BSc Computer Systems Engineering — CGPA 3.57
Oct 2020 – Jun 2024

The Islamia University of Bahawalpur

FSc (Pre-Engineering) — 77.81%
Sep 2018 – Jun 2020

Global Degree College, Peshawar

CERTIFICATIONS

- Google Data Analytics Case Study (Coursera) • Supervised Machine Learning: Regression and Classification
- Unsupervised Learning, Recommenders, Reinforcement Learning • AI Certification (10-month course)
- Introduction to Cybersecurity Tools & Cyberattacks